

Cell Lines and treatments. NB-derived cell lines, SH-SY5Y, SH-EP, SK-N-SH, SKN-BE(2)C, IMR-32 and Chronic Myeloid Leukemia cell line K562 were obtained from LGC Standards-ATCC. CHLA-255 cell line was kindly provided by Prof M. Brenner (Baylor College of Medicine, TX, USA). TET 21/N cell line was kindly provided by Dr M. Schwab (University of Heidelberg, Heidelberg, Germany). NB-derived cell lines LAN-1, SK-N-AS were obtained from SIGMA-ALDRICH while GICAN and ACN from Interlab Cell Line Collection, Banca Biologica and Cell Factory. The LAN-1, ACN, SK-N-AS, SH-EP, SK-N-SH, SKN-BE(2)C, GICAN cell lines were cultured in RPMI 1640 medium (Euroclone, IT), the SH-SY5Y cell line in DMEM medium (Euroclone, IT), and IMR-32 and CHLA-255 in IMDM medium (Euroclone, IT), supplemented with 10% fetal bovine serum (Gibco-Thermo Scientific PA - USA) and 2 mM Gluta-Max (Euroclone, IT) respectively. ACN cell line was also supplemented with 1% non-essential Amino Acid (Euroclone, IT). Cells were maintained in a humidified atmosphere containing 5% CO₂ at 37°C. All cell lines were routinely tested for mycoplasma infection and authenticated by means of short tandem repeat analysis (Eurofins Genomic, Ebersberg, Germany). Where indicated, cell lines were treated with 100 ng/ml IFN γ (Immunological Sciences, IT) or 10 ng/ml Doxycycline (Merck, DE). IDO1 inhibition was performed by the addition of 50nM Epacadostat inhibitor (DBA, IT) or BMS-986205 10nM (Aurogene, IT).

Cell line transfection. Cell lines were stably or transiently transfected with Lipofectamine 2000 (Thermo Scientific, PA - USA) following the manufacturer's instructions. pcDNA3 and pcDNA3-MycN plasmids were used for control and MYCN overexpression (Thermo Scientific PA - USA). pCMV-GFP and pCMV-GFP-IDO1 plasmids were purchased from OriGene, (MD, USA) and used for IDO1 overexpression. Positive cell selection was performed by Geneticin (Thermo Scientific, PA - USA) addition to culture media (400 μ g/ml) every 5 days. FlexiTube GeneSolution scramble and siMYCN oligos were purchased from Qiagen, IT.

Human NK-cell isolation. Human NK cells were isolated from PBMC of healthy donors with the RosetteSep NK-cell enrichment mixture method (Stem-Cell Technologies, IT). NK cells with purity greater than 90% were stimulated with 100 IU/mL of recombinant human IL2 (PeproTech, FR) for 18 hours at 37°C. NK cells were maintained in culture with NK MACS medium supplemented with 10% human serum and 1% NK MACS supplement (Miltenyi Biotech, DE).

Western Blot Analysis. Cell pellets were lysed with RIPA buffer [150 mM NaCl, 1% NP-40, 0.5% sodium deoxycholate, 0.1% SDS, 50 mM Tris–HCl (pH = 8), 1 mM PMSF, 1 mM EGTA, 50 mM NaF, 50 mM Na₃VO₄ and protease inhibitors (Roche, IT)]. Cell lysates were incubated on ice for 20 minutes and clarified by centrifugation at 14,000 rpm for 20 minutes. Cell extracts obtained with RIPA buffer were boiled for 5 minutes at 95°C and analyzed by SDS-PAGE. Samples were transferred onto nitrocellulose membranes (Immunological Science, SIC, IT). Blots were probed with primary antibodies: anti-IDO, anti-MycN and GAPDH (Cell Signaling, Euroclone, IT), washed with PBS and developed with HRP-conjugated rabbit secondary antibodies (Jackson ImmunoResearch, UK).

IDO1 activity. TRP and kynurenine levels were measured with reverse-phase HPLC Agilent Technologies 1200. Briefly, sample aliquots (200 µL) were diluted with 200 µL potassium phosphate buffer (0.05 mol/L pH 6.0) containing the internal standard 3-nitro-L-tyrosine (100 mmol/L). Proteins were precipitated with 50 µL of trichloroacetic acid (2 mol/L). Capped tubes with the precipitate were immediately vortex-mixed and centrifuged for 10 min at 13,000 g. One-hundred-fifty microliters of supernatant were transferred into microvials and placed in the autosampling device. Samples were analyzed using a Kinetex_5-µm C18, LC column 150 X4.6mm (Phenomenex, CA-USA), a double-pump HPLC apparatus (Agilent Technologies, CA-USA) equipped with spectrophotometric and fluorescence detectors. TRP was detected by a fluorescence detector at an excitation wavelength of 285 nm and an emission wavelength of 365 nm. Kynurenine and nitrotyrosine were detected by recording UV absorbance at a wavelength of 360 nm. Elution solvent was as follows: buffer A (potassium phosphate solution (0.015 mol/L, pH 6.4) containing 27 mL of acetonitrile) and Buffer B (acetonitrile). Analysis was carried out at a flow rate of 1 mL/min at 25°C for 12 min. Concentration of components was calculated according to peak heights and was compared with both 3-nitro-L-tyrosine as internal standard and reference curves constructed with L-tryptophan (concentration 10, 20, 30 mmol/L) and KYN (10, 20, 30 mmol/L). The intra-daily coefficient of variation was 1.20% for TRP and 1.25% for kynurenine. Inter-days coefficient was 3.5% for TRP and 3.8% for kynurenine.

RNA isolation and quantitative RT–PCR analysis. Total RNA was isolated using ReliaPrep RNA tissue Miniprep System (Promega, IT) following the manufacturer's instructions. cDNA was synthesized with the GoScript Reverse Transcription system (Promega, IT) following the manufacturer's instructions. The cDNAs were used for

quantitative real-time PCR (qRT–PCR) experiments carried out by QuantStudio Real Time PCR (Thermo Scientific, PA - USA). $\Delta\Delta C_t$ values were normalized with those obtained from the amplification of the endogenous β -actin gene. The following human-specific primers were synthesized by Merck and employed in RT–PCR amplifications:

human IDO1 forward 5'-TCACAGACCACAAGTCAGC2'

human IDO1 Reverse 5'-AGTTGGCAGTAACAGCA-3'

human MycN Forward 5'-CACAAGGCCCTCAGTACCTC-3'

human MycN Reverse 5'-ACCACGTCGATTTCTTCCTC-3'

human β -actin Forward 5'-CCAACCGCGAGAAGATGA-3'

human β -actin Reverse 5'-CCAGAGGCGTACAGGGADAG-3'

ELISA. IFN γ production was quantitated using a commercially available enzyme-linked immunosorbent assay (ELISA; R&D Systems, MN-USA). The limit of detection was 15.6 pg/ml IFN γ .

Immunohistochemistry

Three μ m paraffin-embedded sections were dehydrated, pretreated with avidin block for 15 min followed by a biotin block for other 15 min. Then, slides were incubated with bovine serum albumin (Merck, DE) (BSA) for 30 min and incubated with primary antibody against IDO1 (Santa Cruz Biotechnology, rat monoclonal Ab, 1:250) for 60 min at 37°C and then with secondary antibodies (polyclonal rabbit anti-rat biotinylated Igs Dako, 1:200) for 30 min at 37°C. Chromogenic development was obtained using Fast Red detection kit (Dako Real kit, Agilent, CA-USA). Sections were visualized with a Zeiss AXIO microscope and digital image acquisition performed with attached Zeiss AxioCam MRc5 using the AxioVision 4.8 software.

Chromatin immunoprecipitation assay (ChIP)

Anti-MYCIN antibody (Cell Signaling, Euroclone, IT). Immunoprecipitations with immunoglobulins (Santa Cruz, CA-USA) were performed as negative controls. For quantitative ChIP analysis (ChIP–qPCR), 1 μ l of purified DNA was used for amplification on an Applied Biosystems 7500 Fast Real-Time PCR system (Applied Biosystem SYBR Green; Thermo Scientific, PA-USA). The following human promoter-specific primers were employed in RT–PCR amplifications:

- for e-box sequence

human IDO1 forward 5'-TTGGAAAACTGACAAAAGTCCC-3'

human IDO1 reverse 5'-CAGCCCGGAAACACAACCTGT-3'

- for gas sequence

human IDO1 forward 5'-TTAGCCTCATGAATCATGTAG-3'

human IDO1 reverse 5'-ATATGGCTTTCGTTACAGTCT-3'

Mutagenesis

Mutagenesis of mIDO1 promoter (Murine IDO1 promoter) gently provided from Prof Fallarino (University of Perugia, Perugia, IT) was performed using the QuikChange Mutagenesis kit (Stratagene, CA-USA), altering the canonical e-box sequence CACGTG in TATATG and in CCTGAG. Sequencing was realized by Genechron (Rome, IT).

Luciferase assay

Cell extracts were prepared and assayed for luciferase activity according to the manufacturer's instructions (Promega, IT) using the GloMax Luminometer (Promega, IT). Total protein quantification in the extracts was determined by Bradford assay and luciferase activity of an equal amount of proteins was determined.